

Name: _____

Solving Radical Equations

Day 1 Practice: Equations That Create Quadratics

Problems 1-8 Solve the radical equation. Check for extraneous solutions.

1. $\sqrt{3x} = x - 6$

2. $\sqrt{x + 12} = x$

3. $(3x - 2)^{\frac{1}{2}} = x - 2$

4. $(8x + 20)^{\frac{1}{2}} = x$

5. $\sqrt{x - 3} = x - 5$

6. $\sqrt{5x + 4} = x + 2$

7. $\sqrt{7x + 4} = x + 2$

8. $\sqrt{2x - 5} = x - 4$

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Day 2 Practice: Two Radicals – Squaring Twice

Problems 1-6 Solve the radical equation. Check for extraneous solutions.

1. $\sqrt{x+7} + 1 = (x+16)^{\frac{1}{2}}$

2. $\sqrt{3x+4} - 2 = \sqrt{x}$

3. $\sqrt{x+5} - \sqrt{x-3} = 2$

4. $\sqrt{3x+1} + 1 = (5x)^{\frac{1}{2}}$

5. $\sqrt{x+9} = \sqrt{x-3} + 2$

6. $(2x+6)^{\frac{1}{2}} - (x+4)^{\frac{1}{2}} = 1$

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Day 3 Practice: Mixed Review & Applications

Problems 1-6 Identify the type, solve, and check for extraneous solutions.

1. $(4x + 1)^{\frac{1}{2}} = (2x + 9)^{\frac{1}{2}}$

Type: _____

2. $\sqrt{3x + 4} = x$

Type: _____

3. $\sqrt{x + 3} - 1 = \sqrt{x - 2}$

Type: _____

4. $\sqrt{x + 7} = x - 5$

Type: _____

5. $\sqrt{x + 3} + 2 = \sqrt{x + 11}$

Type: _____

6. $\sqrt{6x + 1} = x + 1$

Type: _____

7. $3\sqrt{x} - 2x = -5$

8. $4\sqrt{x - 3} - x = -3$

Problem 9 (Extended Response – Show all work)

The speed of a tidal wave, s , in hundreds of miles per hour, can be modeled by the equation

$$s = \sqrt{2t + 6} - t$$

where t represents the time from its origin in hours.

Part A: Algebraically determine the time when $s = 0$.

Part B: How much faster was the tidal wave traveling after 1 hour than after 3 hours, to the nearest mile per hour? Justify your answer.

Example 4 The time t (in seconds) it takes an object to fall from a height h (in feet) is given by the formula:

$$t = \sqrt{\frac{h}{16}}$$

If an object takes 2.5 seconds to hit the ground, from what height was it dropped?